

1. The endpoints of the directed line segment NW are $N(-14,9)$ and $W(10,-6)$. Determine point D such that line segment NW is partitioned in a ratio of 1:2.

work	
Point D	$(-6, 4)$

2. The center of a circle H is at $(-3,-2)$ and has a radius that is $\sqrt{10}$ units. Select all of the points that are on circle H . Show your work.

- ☐ $(-5, 0)$
- ☒ $(-4, -5)$
- ☐ $(-2, -3)$
- ☒ $(-2, -5)$
- ☒ $(0, -1)$
- ☐ $(8, 7)$

work

3. The vertices of quadrilateral $DMYW$ are shown.

$D(10,3)$

$M(3,-2)$

$W(0,5)$

$Y(-7,0)$

Select all the classifications of quadrilateral $DMYW$. Show your work.

work

- ☐ kite
- ☐ square
- ☐ rhombus

- ☐ rectangle
- ☒ trapezoid
- ☒ parallelogram

4. The vertices of $\triangle THB$ are shown.

$B(-5, -2)$

$H(3, -4)$

$T(-3, 6)$

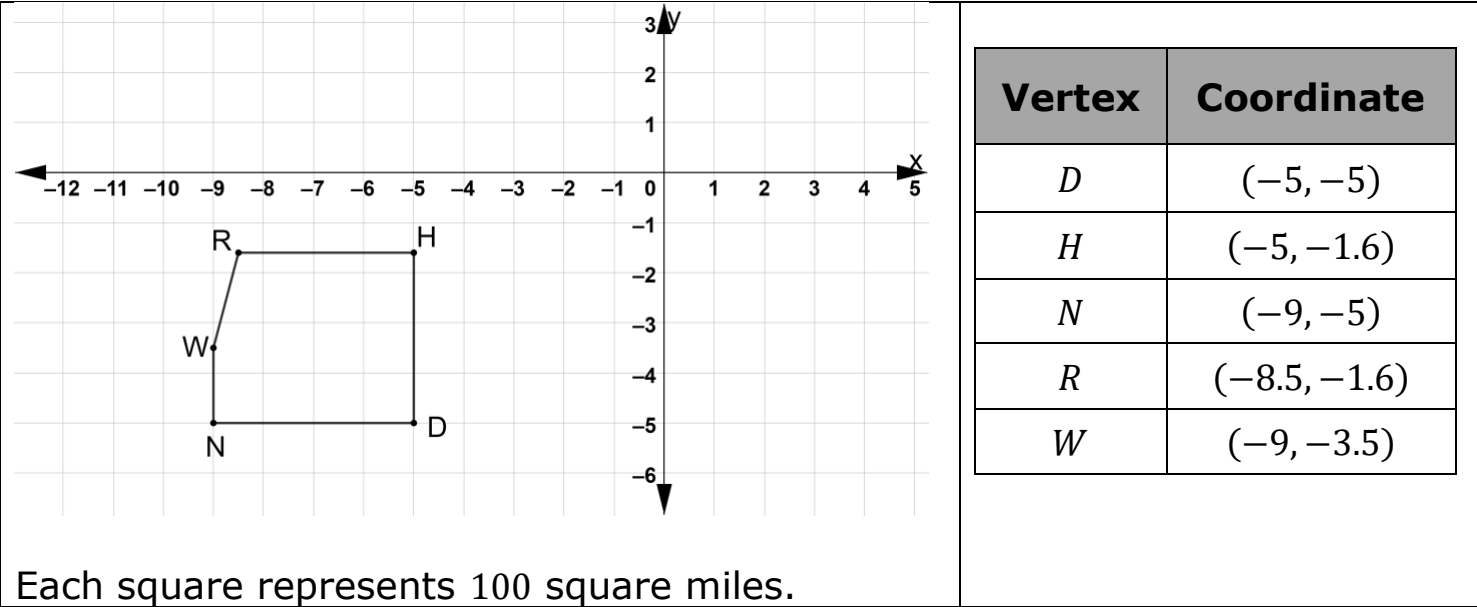
a. Choose one word from each column to describe $\triangle THB$. Show your work.

work	
☐ scalene ☒ isosceles ☐ equilateral	☒ right ☐ acute ☐ obtuse

b. Determine the coordinates of the centroid of $\triangle THB$. Show your work.

work	
Centroid	$\left(-1\frac{2}{3}, 0\right)$ or $\left(-\frac{5}{3}, 0\right)$

5. The Florida Reef is the only coral reef in North America. It stretches along Martin County to the Dry Tortugas National Park. The Florida Reef contains the oldest known coral colony, which is over 300 years old. An ocean scientist was studying the part of the protected area that extends from the Dry Tortugas to Key West. She drew the area on the coordinate grid, as shown.



Determine the area, in square miles, of the area that the ocean scientist is studying.

work	
Area	1312.5 mi ²